Ayan Chattopadhyay

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OBJECTIVE

A computational quantum theorist by training, with 8+ years of scientific computing experience in developing mathematical models to study complex systems & implementing robust codes (Python OOP: 6+ years) for their analyses, using advanced machine learning & optimization techniques. Looking to utilize these mathematical & programming skills to solve challenging problems in quantitative finance.

EDUCATION

Princeton University, *Princeton, NJ*
PhD in Theoretical & Quantum Chemistry *Advisor: Prof. Herschel Rabitz* *Sep. 2014 – Sep. 2020*
RESEARCH AREAS: ML & Optimization: Numerical Algorithms for Quantum Applications, Linear Algebra
GPA (Major): 3.70 / 4.00 (Overall): 3.52 / 4.00

Indian Institute of Technology, Kharagpur, *Kharagpur, India*
Masters & Bachelors (Honors) in Chemistry *Advisor: Prof. P. K. Chattaraj* *Aug. 2009 – Apr. 2014*
GPA (Major): 8.89 / 10.00 (Overall): 8.72 / 10.00

TECHNICAL SKILLS

- **Mathematics** : Probability Theory, Linear Algebra & Matrix Theory, Optimization (Convex & Nonlinear), Numerical Techniques, Mathematical Modeling.
- **Machine Learning** : Linear Models, Clustering Algorithms, Dimensionality Reduction, Neural Networks, Genetic Algorithms, Gradient Algorithms.
- **Programming** : Python(Proficient), C, C++, R, Mathematica, Parallel Computing in OpenMP.
- **Data Science Tools** : Tensorflow, Keras, Scikit-Learn, Pandas.

RELEVANT COURSES

- Fundamentals of Quantitative Modeling (The Wharton School) *Coursera Certificate*
- Trading Algorithms (Indian School of Business) *Coursera Certificate*
- Advanced Trading Algorithms (Indian School of Business) *Coursera Certificate*
- Linear System Theory [Linear Algebra and Machine Learning] *Princeton U.*
- Applied Algebra [Group Theory & Representation Theory] *Princeton U.*
- Mathematical Methods for Engineering Analysis [Engineering Mathematics] *Princeton U.*
- Optimal Control & Estimation [Control Theory & Optimization] *Princeton U.*
- Numerical Solutions of ODEs & PDEs [Advanced Numerical Methods] *I.I.T. Kharagpur*
- Industrial Economics *I.I.T. Kharagpur*
- Introduction to Financial Management *I.I.T. Kharagpur*

EXPERIENCE

Graduate Research Assistant, *Princeton University*
Quantum Control Research group *Sep. 2014 – Present.*

- Worked in the Quantum Control Research group as a theoretical and computational scientist.
- Derived and developed mathematical and machine learning models from scratch to study quantum dynamics of complex physical systems.
- Implemented models in *Python* and *C* predicting important parameters to support experiments in the group.
- Lead the complete development and analysis of a novel theoretical method in the area of nonlinear spectroscopy, utilizing linear algebra, complex analysis and convex optimization theory.

Graduate Teaching Assistant, *Princeton University*
General Chemistry I & II; Advanced General Chemistry *Fall 2015, Fall 2017, Spring 2018*

- Guided approximately 85 students overall with experiments and graded exams and assignments.

RESEARCH PROJECTS & PUBLICATIONS

Theory of nonlinear optical spectroscopy with Optical Frequency Combs *PRINCETON U., NJ*

- Derived complete theoretical model utilizing complex analysis & quantum perturbation theory.
- Performed complete theoretical analysis utilizing linear algebra & matrix decomposition tools.
- Implemented the theory in *Python, C, OpenMP & NLopt* (nonlinear optimization library in C). Verified complex Cauchy integrals in *Mathematica*. Implemented Gradient Descent (in C) and Genetic Algorithm in *DEAP* (*Python*).
- *Publication*: "Efficient non-linear optical spectroscopy with multiple high-frequency modulated optical frequency combs" *Optical Society of America, Frontiers in Optics*. (Finalist: Emil Wolf Outstanding Paper Competition)
- (*Manuscript available*) "Theory for optical-frequency-comb-based nonlinear spectroscopy"
- (*In preparation*) "Discrimination of spectrally similar sensors with frequency-comb-based nonlinear spectroscopy"

Raman spectroscopy assisted discrimination of quantum systems *PRINCETON U., NJ*

- Derived a theoretical model utilizing quantum dynamics and control theory to guide related laser experiments.
- Implemented the theory in *Python, C & OpenMP*, fit related experimental data with optimization routines in C.
- (*Manuscript available*) "Reduction of spectral cross-talk in optogenetics with Raman assisted optical control"

Master's Thesis: Zero quantum potential limit of a driven double-well oscillator *I.I.T. KHARAGPUR, INDIA*

- Derived a theoretical model using Bohmian quantum dynamics & implemented it using numerical techniques in *FORTRAN*. Theoretically analyzed chaos in simple quantum systems.

Coupled Cluster Theory of Raman Optical Activity (*Summer '13*) *VIRGINIA TECH, VA*

- Derived complete mathematical model using quantum perturbation theory & implemented it in C++.

Path Integral Monte Carlo & Vibronic Coupling Theory of non-adiabatic systems *U. WATERLOO, ON*

- Implemented a quantum variant of Monte Carlo algorithm in FORTRAN to study special quantum systems.

AWARDS

Program in Plasma Science & Technology, *Princeton, NJ* *May. 2018 – Aug. 2019*

Robert S. Hilbert Memorial Award, Optical Society of America, *Washington D.C.* *Sep. 2018*

Jean Bennett Memorial Award, Optical Society of America (declined), *Washington D.C.* *Sep. 2018*

Taylor Fellowship, *Princeton, NJ* *Sep. 2017 – May. 2018*

INSPIRE Scholarship, *IIT Kharagpur, WB, India* *Aug. 2009 – May. 2014*

TEST SCORES

- **GRE:** [330/340] (Quant *170/170*; Verbal *160/170*; Writing *4.0/6.0*) *2013*
- **TOEFL iBT:** [111/120] *2013*